Quantification: 682_z29
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1885	1629	856.283	px^2
major_axis_length_px	lacuna shape	79.6556	78.7239	19.8072	px
minor_axis_length_px	lacuna shape	30.5623	31.1018	8.0704	px
aspect_ratio	lacuna shape	2.7208	2.6218	0.7334	unitless
eccentricity	lacuna shape	0.9094	0.9241	0.0653	unitless
solidity	lacuna shape	0.8664	0.8806	0.0554	unitless
roots_count	headline	9.1818	10	2.892	count
ring_length_r30_px	headline	359.909	376	137.096	px
ring_length_r60_px	reported	843.273	939	320.125	px
owned_length_px	ownership-dependent	2791.76	1579.88	2247.3	px
edge_count	ownership-dependent	123.091	68	105.714	count
mean_edge_length_px	ownership-dependent	23.5147	23.1351	1.8123	px
perimeter_px	normalised	204.585	214.167	56.737	px
ring_area_r30_px2	normalised	8010.18	8595	2022.59	px^2
ring_area_r60_px2	normalised	19953	20090	4908.42	px^2
in_frame_fraction_r30	normalised	0.9943	1	0.0188	unitless
in_frame_fraction_r60	normalised	0.981	1	0.0583	unitless
ring_density_r30	normalised	0.0440777	0.0437464	0.00947067	px^-1
ring_density_r60	normalised	0.0414318	0.0433264	0.00835928	px^-1
roots_per_100px_perimeter	normalised	4.5374	4.6761	1.04	per 100 px
ring_attached_length_r30_px	network v2	335.091	358	129.292	px
ring_attached_length_r60_px	network v2	730	739	321.633	px
ring_length_w_r30_px	network v2	405.546	420.735	151.058	px
ring_length_w_r60_px	network v2	952.076	1052.85	361.247	px
sholl_crossings_r10	network v2	8.9091	10	2.8794	count
sholl_crossings_r20	network v2	11.0909	12	4.1822	count
sholl_crossings_r30	network v2	12.3636	12	4.4333	count
loop_count	loops, report only	13.0909	3	17.8463	count
loop_edge_share	loops, report only	0.2343	0.1786	0.2321	share
ring_width_mean_r30_px	width	6.9213	6.7132	0.6735	px

Per field
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measure	value	units
analysed_area_px2	1.02335e+06	px^2
total_skeleton_length_px	36867	px
canalicular_length_density_per_px	0.0360258	px^-1
junction_count	824	count
junction_density_per_px2	0.0008052	px^-2
lacuna_count	13	count

lacunae_per_px2	1.27e-05	px^-2
skeleton_component_count	230	count
mean_component_length_px	160.291	px
median_component_length_px	20.5	px
field_length_density_w_per_px	0.0407128	px^-1
field_density_without_flagged_per_px	0.0371543	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	443	px
unattached_component_count	28	count
canalicular_length_density_attached_per_px	0.0360258	px^-1
field_length_density_with_unattached_per_px	0.0364587	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	166	count
loop_edge_share	0.3512	share
edge_count	1808	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7447	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	35210	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
4	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
5	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
6	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
7	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
8	663	61.7115	15.8655	3.8897	0.9664	0.7408	-1.1603	728.596	861.06	False
9	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
10	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
11	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
12	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False
13								830.565	75.5867	False

Network per lacuna

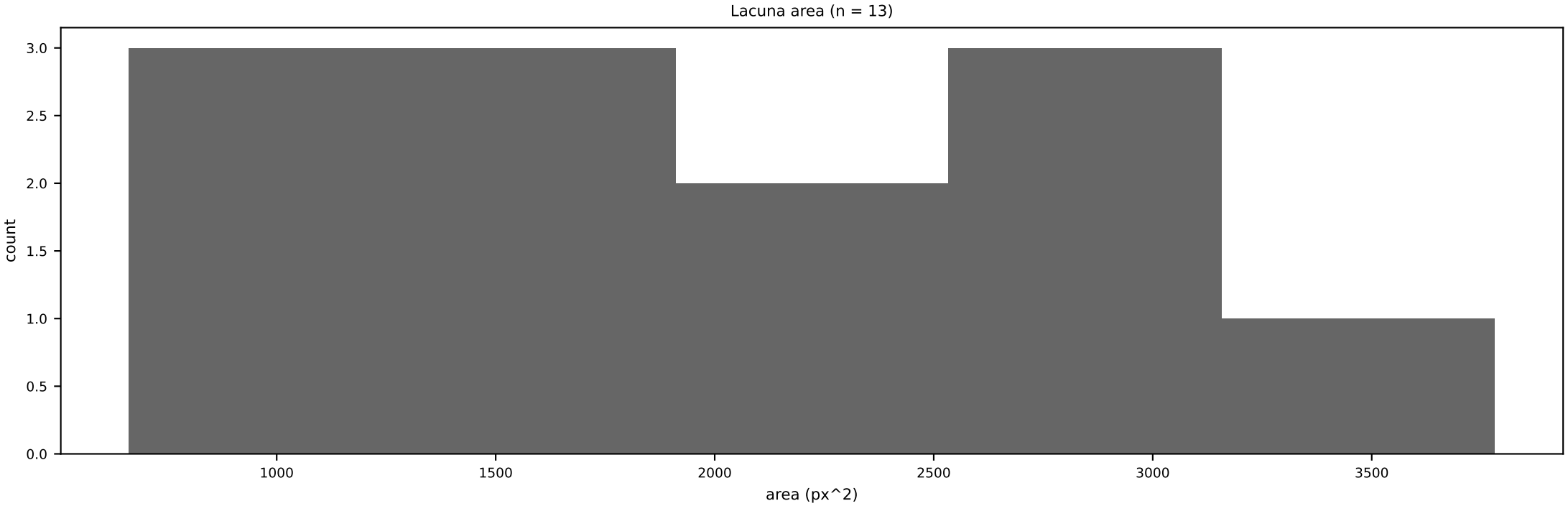
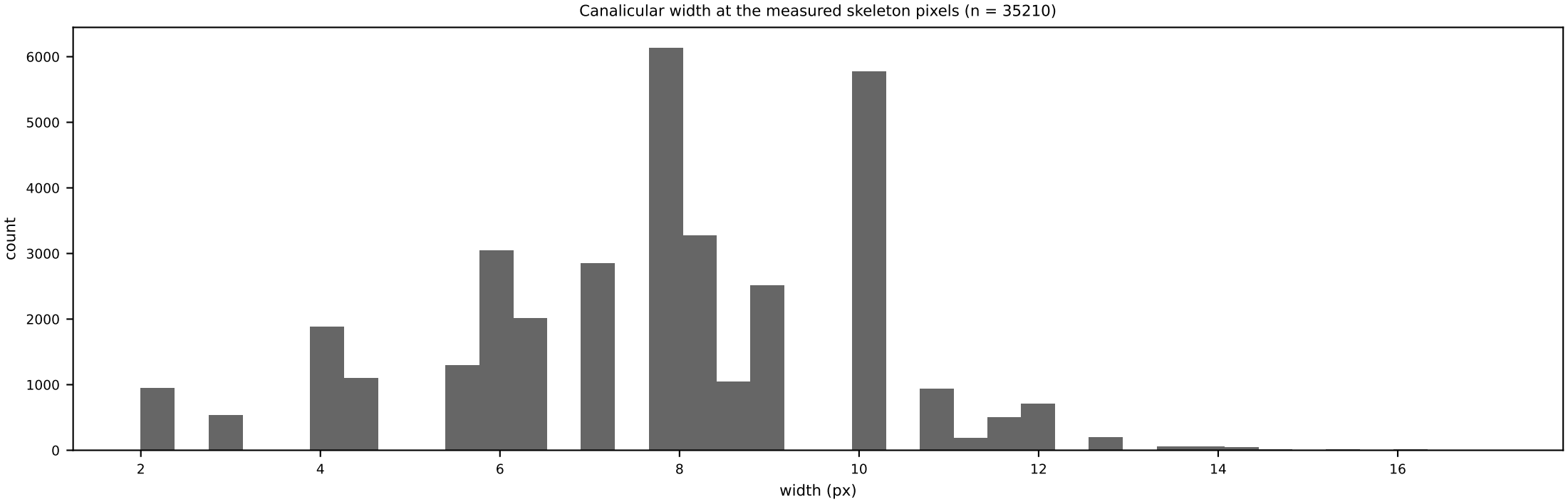
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	192	359	1688.02	91	18.5496	
2	14	556	1343	6503.61	282	23.0624	6.9555
3	11	475	1301	6273.37	317	19.7898	7.5257
4	11	468	1010	4719.55	204	23.1351	6.4148

5	11	510	993	3670.21	159	23.0831	6.8081
6	11	392	836	1236.62	60	20.6103	7.4798
7	8	376	955	1369.39	56	24.4534	6.2347
8	9	261	717	3322.21	135	24.609	6.5368
9	5	191	480	626.831	24	26.1179	8.6113
10	5	240	619	1567.94	68	23.0579	6.4606
11	11	445	939	1579.88	62	25.482	6.7132
12	10	249	438	563.245	23	24.4889	7.2273
13	6	188	481	513.162	24	21.3817	6.6465

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0342429	0.0259787	1.5262
2	229.522	9151	23871	1	0.999	0.0607584	0.0562607	6.0996
3	309.279	11331	28206	1	1	0.0419204	0.0461249	3.5567
4	235.238	9387	21888	1	1	0.0498562	0.046144	4.6761
5	214.167	8637	20090	1	1	0.0590483	0.0494276	5.1362
6	322.836	9940	22198	0.8402	0.7724	0.0394366	0.0376611	3.4073
7	229.48	8595	22042	1	1	0.0437464	0.0433264	3.4861
8	156.853	6922	19429	1	1	0.0377059	0.0369036	5.7379
9	161.817	5491	13376	1	1	0.0347842	0.0358852	3.0899
10	147.196	6914	19480	1	1	0.0347122	0.0317762	3.3968
11	254.693	9728	23664	1	1	0.0457442	0.0396805	4.3189
12	202.693	7711	16371	0.9376	0.8055	0.0322915	0.0267546	4.9336
13	109.498	4245	11066	1	0.9865	0.0442874	0.0434665	5.4796

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	150	219.51	395.664	2	6	6
2	548	1291	617.718	1522.9	12	16	18
3	444	1224	542.375	1471.6	12	14	17
4	439	845	512.948	1116.42	12	15	16
5	458	943	574.961	1119.85	11	13	15
6	379	771	440.563	939.968	10	12	12
7	358	739	420.735	1076.54	10	12	12
8	216	532	304.492	835.222	6	8	11
9	191	450	221.823	534.09	5	7	8
10	229	509	273.451	713.754	7	8	9
11	388	744	500.718	1052.85	11	17	17
12	239	361	281.451	494.031	6	6	7
13	176	392	210.338	535.576	6	6	6

682_z29: distributions, pre-validation, px